

Assignment 6: CST8912 – Cloud Solution Architecture

Graded Lab Activity #6

Introduction: Azure Private Link, a secure connectivity option for private network access to Azure services, is the main emphasis of this lab. By guaranteeing that data traffic stays inside the Azure virtual network (VNet) and is not exposed to the public internet, Azure Private Link streamlines network architecture. By providing a private, direct connection, Private Link enables customers to safely access Azure PaaS services, Microsoft partner services, or even custom services developed by their company.

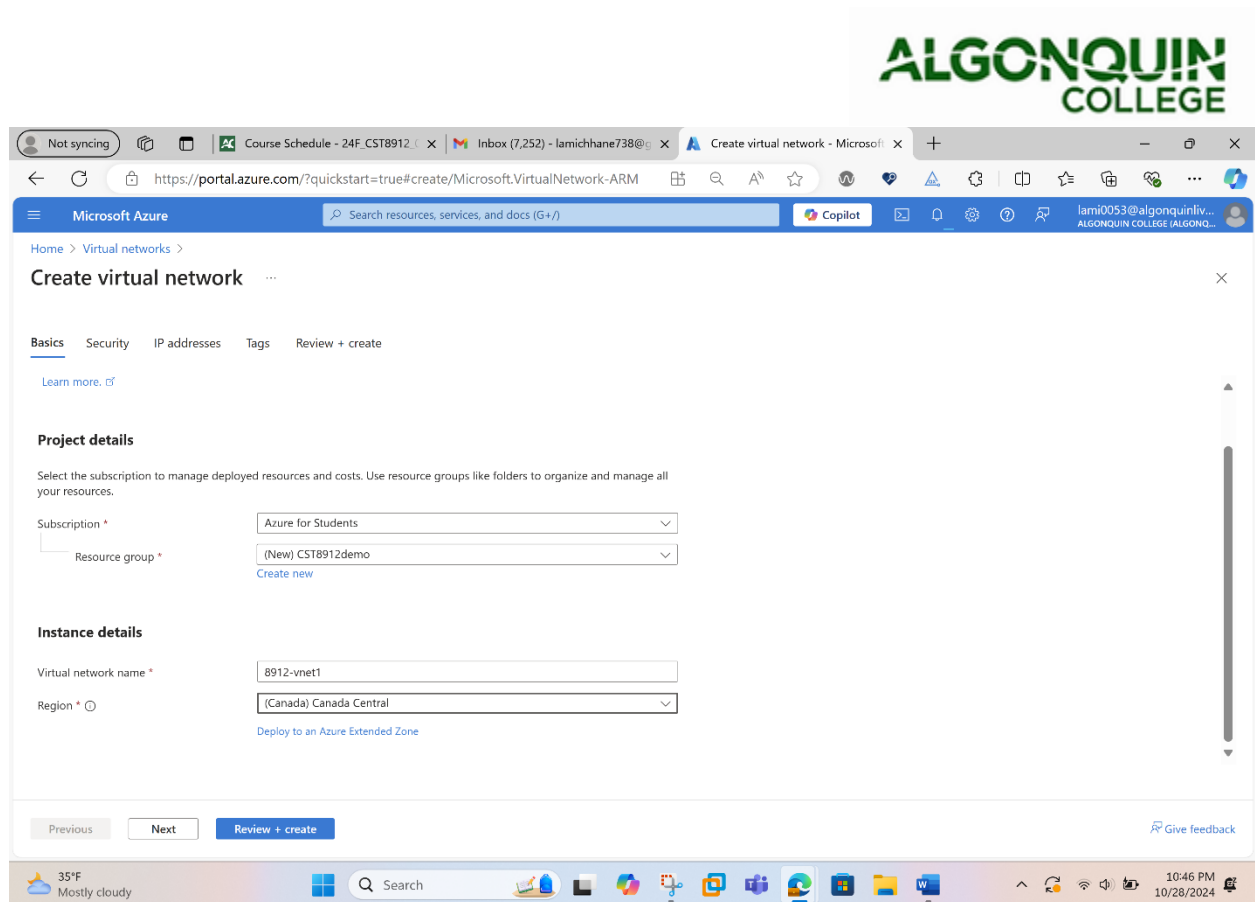
Objective: Enabling you to construct, configure, and test an Azure Private Link and Private Endpoint is the aim of this experiment. By the conclusion, you will know how to safely link Azure services to a virtual network, satisfying regulatory requirements, lowering the danger of internet exposure, and guaranteeing confidential communication between your business's bespoke Azure services and clients.

Step 1: Create a Virtual network and Bastion Host

1. In the portal, we searched for and select Virtual networks. On the Virtual networks page, we selected + Create.

On the Basics tab of Create virtual network, we entered the following information:

Setting	Value
Subscription	Azure for students
Resource Group	CST8912demo
Name	8912-vnet1
Region	Canada Central



2. Then we select Next to proceed to the Security tab. Select Enable Bastion in the Azure Bastion section of the Security tab. Enter or select the following information in Azure Bastion:

Setting	Value
Azure Bastion host name	Enter Bastion8912
Azure Bastion public IP address	Select Create a public IP address . Enter public-ip in Name. Select OK .

The screenshot shows the Microsoft Azure portal interface for creating a virtual network. The 'Security' tab is selected, displaying options for Azure Bastion and Azure Firewall. A modal window titled 'Add a public IP address' is open, showing the following details:

- Name ***: public-ip
- SKU**: Standard (selected)
- Assignment**: Static (selected)

The background form shows the 'Create virtual network' wizard with the following fields:

- Enable Azure Bastion**: ☒
- Azure Bastion host name**: Bastion8912
- Azure Bastion public IP address ***: (New) 8912-vnet1-bastion
- Enable Azure Firewall**: ☐

Navigation buttons at the bottom include 'Previous', 'Next', and 'Review + create'. The 'Review + create' button is highlighted in blue.

3. Select Next to proceed to the IP Addresses tab. In the address space box in Subnets, select the default subnet. In Edit subnet, enter or select the following information:

Setting	Value
Subnet template	Leave the default Default .
Name	Subnet-1-8912
IPv4 address range	Leave 10.0.0.0/16.
Starting address	Leave the default of 10.0.0.0.
Subnet size	Leave the default of /24(256 addresses).

The screenshot displays the Microsoft Azure portal interface. The main navigation bar at the top shows the user is logged in as 'lami0053@algonquiniv...' and the current page is 'Edit subnet - Microsoft Azure'. The breadcrumb trail indicates the path: Home > Virtual networks > Create virtual network.

The 'Create virtual network' page is active, showing the 'IP addresses' tab. It includes a section for 'Add a subnet' with a table listing subnets:

Subnets	IP address range	Size
default	10.0.0.0 - 10.0.0.255	/24 (256 addresses)
AzureBastionSubnet	10.0.1.0 - 10.0.1.63	/26 (64 addresses)

Below the table, there is a button 'Add IPv4 address space' and a dropdown menu.

The 'Edit subnet' modal is open, showing configuration options for the 'default' subnet. The 'Subnet purpose' is set to 'Default'. The 'Name' is 'default'. Under the 'IPv4' section, 'Include an IPv4 address space' is checked. The 'IPv4 address range' is '10.0.0.0/16', with a detailed range of '10.0.0.0 - 10.0.255.255'. The 'Starting address' is '10.0.0.0'. The 'Size' is '/24 (256 addresses)'. The 'Subnet address range' is '10.0.0.0 - 10.0.0.255'. Under the 'IPv6' section, 'Include an IPv6 address space' is unchecked, with a note: 'This virtual network has no IPv6 address ranges.' The 'Private subnet' section provides information about security and outbound access.

At the bottom of the modal, there are 'Save' and 'Cancel' buttons, and a 'Give feedback' link.

Next step is to select save and the create and review when the validation passes.

Not syncing | Course Schedule - 24F, CST8912 | Inbox (7,252) - lamichhane738@ | Edit subnet - Microsoft Azure

https://portal.azure.com/?quickstart=true#create/Microsoft.VirtualNetwork-ARM

Microsoft Azure | Search resources, services, and docs (G+)

Home > Virtual networks >

Create virtual network

Basics Security IP addresses Tags Review + create

Virtual network address space into smaller ranges for use by your applications. When you use a virtual network, you assign the resource an IP address from the subnet. [Learn more](#)

+ Add a subnet

Subnets	IP address range	Size
default	10.0.0.0 - 10.0.255.255	/24 (256 addresses)
AzureBastionSubnet	10.0.1.0 - 10.0.1.63	/26 (64 addresses)

Add IPv4 address space

Previous Next Review + create

Edit subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. [Learn more](#)

Subnet purpose: Default

Name: Subnet-1-8912

IPv4

Include an IPv4 address space: ☒

IPv4 address range: 10.0.0.0 - 10.0.255.255

Starting address: 10.0.0.0

Size: /24 (256 addresses)

Subnet address range: 10.0.0.0 - 10.0.0.255

IPv6

Include an IPv6 address space: ☐ This virtual network has no IPv6 address ranges.

Private subnet

Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual machines to access the internet, it is necessary to explicitly grant outbound access. A NAT gateway is the recommended way to provide outbound connectivity for virtual machines in the subnet. [Learn more](#)

Save Cancel Give feedback

35°F Mostly cloudy | Search | 10:49 PM 10/28/2024

Not syncing | Course Schedule - 24F, CST8912 | Inbox (7,252) - lamichhane738@ | Create virtual network - Microsoft Azure

https://portal.azure.com/?quickstart=true#create/Microsoft.VirtualNetwork-ARM

Microsoft Azure | Search resources, services, and docs (G+)

Home > Virtual networks >

Create virtual network

Basics Security IP addresses Tags Review + create

[View automation template](#)

Basics

Subscription	Azure for Students
Resource Group	CST8912demo
Name	8912-vnet1
Region	Canada Central

Security

Azure Bastion	Enabled
- Name	(New) Bastion8912
- Public IP Address	(New) public-ip
Azure Firewall	Disabled
Azure DDoS Network Protection	Disabled

IP addresses

Address space	10.0.0.0/16 (65,536 addresses)
---------------	--------------------------------

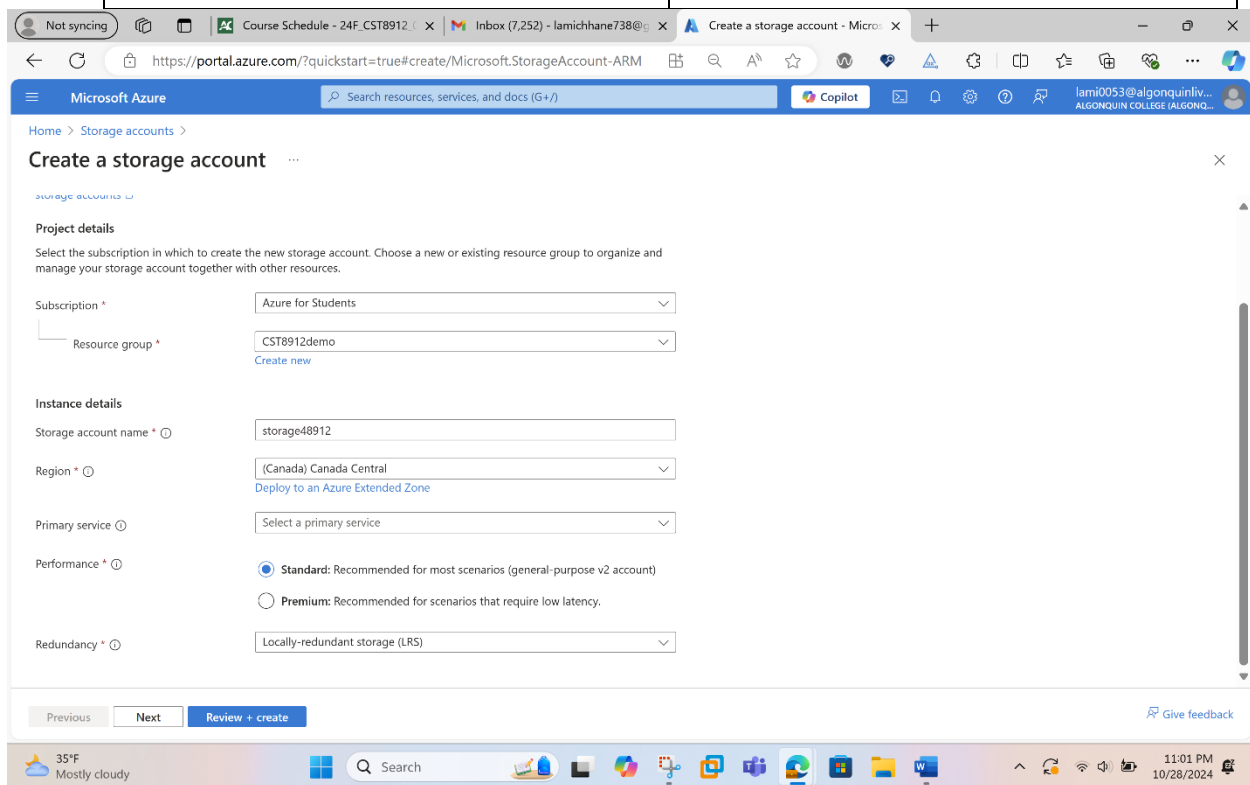
Previous Next Create Give feedback

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Step 2: Creation of a Storage Account

In the search box at the top of the portal, we entered Storage account. Selected Storage accounts in the search results and then selected + Create. Then in the Basics tab to Create a storage account we entered or select the following information:

Setting	Value
Subscription	Azure for students
Resource Group	CST8912demo
Storage Account Name	Storage1-8912
Location	Canada Central
Performance	Standard
Redundancy	Local Redundant Storage (LRS)



Not syncing

Course Schedule - 24F_CST8912... | Inbox (7,252) - lamichhane738@... | Create a storage account - Micro...

https://portal.azure.com/?quickstart=true#create/Microsoft.StorageAccount-ARM

Microsoft Azure | Search resources, services, and docs (G+)

Home > Storage accounts >

Create a storage account

storage accounts

Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription *

Resource group * [Create new](#)

Instance details

Storage account name *

Region * [Deploy to an Azure Extended Zone](#)

Primary service

Performance * ☒ Standard: Recommended for most scenarios (general-purpose v2 account)
☐ Premium: Recommended for scenarios that require low latency.

Redundancy *

[Previous](#) [Next](#) [Review + create](#) [Give feedback](#)

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Finally select review and select create.

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https://portal.azure.com/?quickstart=true#create/Microsoft.StorageAccount-ARM

Microsoft Azure | Search resources, services, and docs (G+)

Home > Storage accounts >

Create a storage account

Basics | Advanced | Networking | Data protection | Encryption | Tags | **Review + create**

[View automation template](#)

Basics

Subscription	Azure for Students
Resource group	CST8912demo
Location	Canada Central
Storage account name	storage48912
Primary service	
Performance	Standard
Replication	Locally-redundant storage (LRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot
Enable large file shares	Enabled

Previous | Next | **Create**

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The overview page of storage looks like this:

Not syncing | Course Schedule - 24F_CST8912... | Inbox (7,252) - lamichhane738@... | storage48912 - Microsoft Azure

https://portal.azure.com/?quickstart=true#@AlgonquinLivecom.onmicrosoft.com...

Microsoft Azure | Search resources, services, and docs (G+)

Home > Storage accounts >

Storage accounts

Algonquin College (AlgonquinLivecom.onmicrosoft.com)

+ Create | Restore | ...

Filter for any field...

Name
storage48912

Overview

- Activity log
- Tags
- Diagnose and solve problems
- Access Control (IAM)
- Data migration
- Events
- Storage browser
- Storage Mover
- Partner solutions
- Data storage
 - Containers
 - File shares
 - Queues
 - Tables
- Security + networking
 - Networking
 - Front Door and CDN

storage48912

Upload | Open in Explorer | Delete | Move | Refresh | Open in mobile | CLI / PS

Essentials

Resource group	(move) CST8912demo
Location	canadacentral
Subscription	(move) Azure for Students
Subscription ID	469222a6-9aad-44fa-b0b8-12052d695b23
Disk state	Available
Tags	(edit) Add tags

Properties | Monitoring | Capabilities (7) | Recommendations (0) | Tutorials | Tools + SDKs

Blob service	
Hierarchical namespace	Disabled
Default access tier	Hot
Blob anonymous access	Disabled
Blob soft delete	Enabled (7 days)

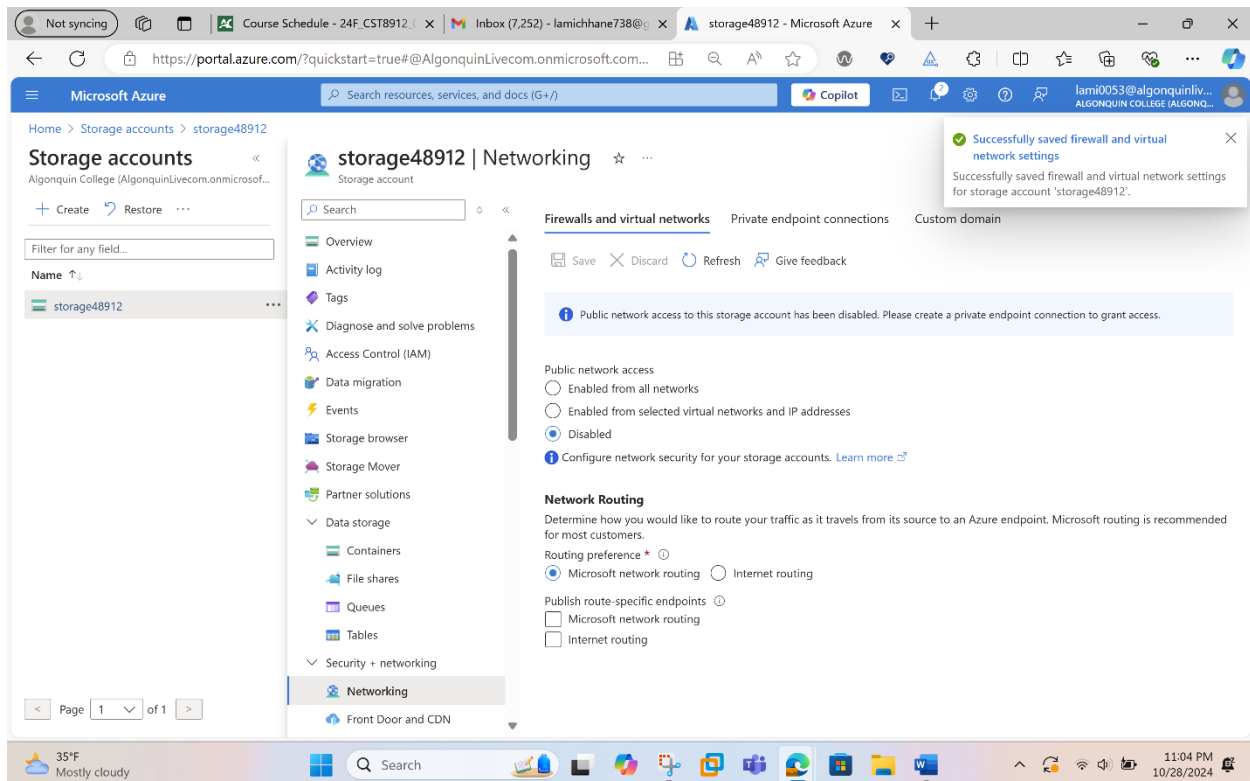
Security	
Require secure transfer for REST API operations	Enabled
Storage account key access	Enabled
Minimum TLS version	Version 1.2
Infrastructure encryption	Disabled

Page 1 of 1

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Step 3: Disable public access to storage account

In the search box at the top of the portal, enter Storage account. Select Storage accounts in the search results. Select storage1 or the name of your existing storage account. In Security + networking, select Networking. In the Firewalls and Virtual Networks tab in Public network access, select Disabled and then finally select Save.

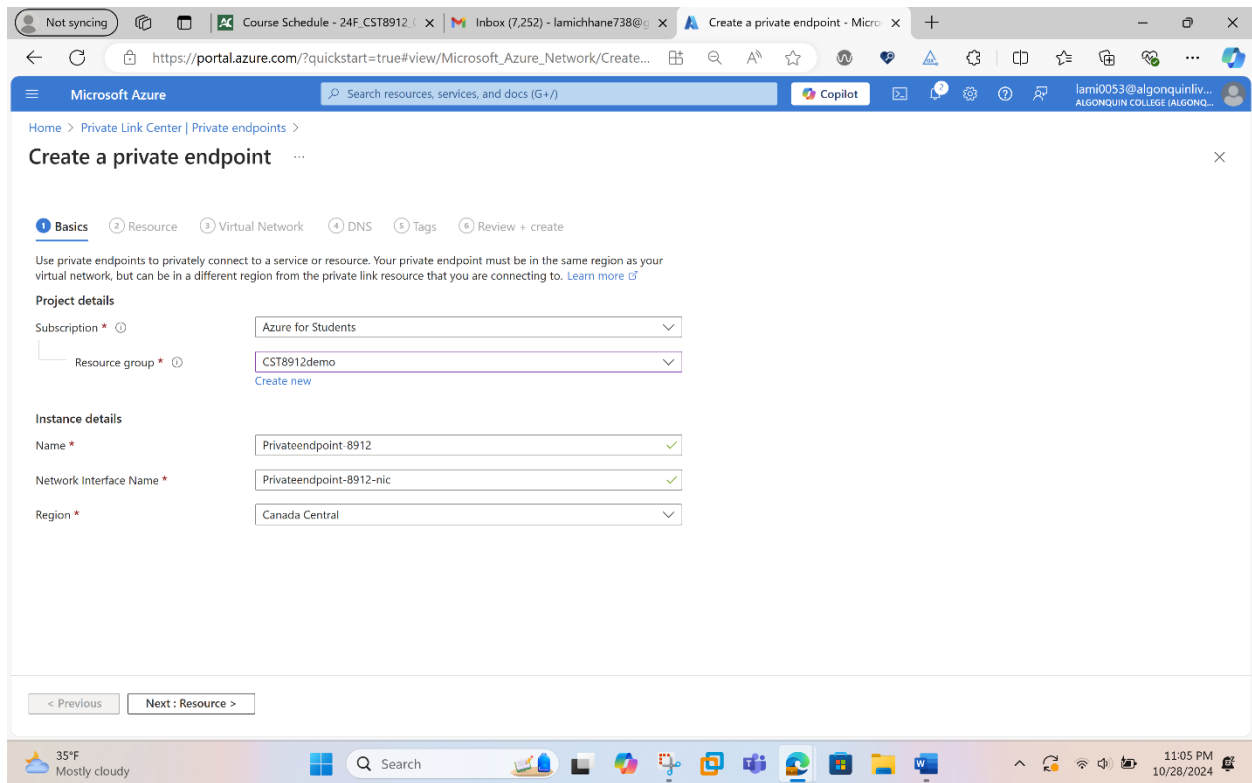


Step 4: Create private endpoint

In the search box at the top of the portal, enter Private endpoint. Select Private endpoints.

Select + Create in Private endpoints and in the Basics tab of Create a private endpoint, enter or select the following information.

Setting	Value
Subscription	Azure for students
Resource Group	CST8912demo
Name	Privateendpoint-8912
Network Interface Name	Leave the default of private-endpoint-nic .
Region	Canada Central



Home > Private Link Center | Private endpoints >

Create a private endpoint

1 Basics 2 Resource 3 Virtual Network 4 DNS 5 Tags 6 Review + create

Use private endpoints to privately connect to a service or resource. Your private endpoint must be in the same region as your virtual network, but can be in a different region from the private link resource that you are connecting to. [Learn more](#)

Project details

Subscription * Azure for Students

Resource group * CST8912demo [Create new](#)

Instance details

Name * Privateendpoint-8912 ✓

Network Interface Name * Privateendpoint-8912-nic ✓

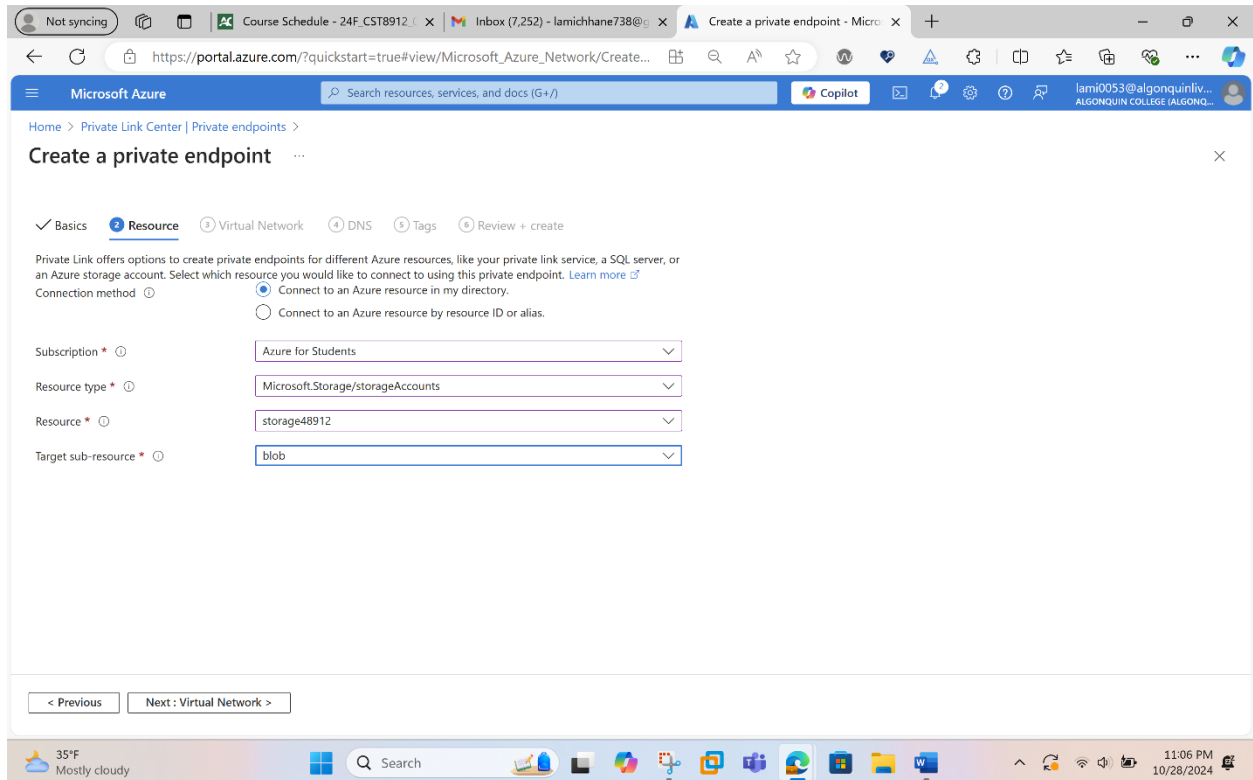
Region * Canada Central

< Previous Next: Resource >

Select Next:Resource

1. In the Resource pane, enter or select the following information.

Setting	Value
Connection method	Leave the default of Connect to an Azure resource in my directory.
Subscription	Select your subscription.
Resource type	Select Microsoft.Storage/storageAccounts.
Resource	Select Storage1-8912 or your storage account.
Target subresource	Select blob.



Home > Private Link Center | Private endpoints >

Create a private endpoint

✓ Basics **Resource** ③ Virtual Network ④ DNS ⑤ Tags ⑥ Review + create

Private Link offers options to create private endpoints for different Azure resources, like your private link service, a SQL server, or an Azure storage account. Select which resource you would like to connect to using this private endpoint. [Learn more](#)

Connection method

☒ Connect to an Azure resource in my directory.

☐ Connect to an Azure resource by resource ID or alias.

Subscription *

Resource type *

Resource *

Target sub-resource *

< Previous Next : Virtual Network >

2. Select Next: Virtual Network. In Virtual Network, enter or select the following information.

Setting	Value
Virtual network	Select 8912-vnet (CST8912demo).
Subnet	Select Subnet-1-8912.
Network policy for private endpoints	Select edit to apply Network policy for private endpoints. In Edit subnet network policy, select the checkbox next to Network security groups and Route Tables in the Network policies setting for all private endpoints in this subnet pull-down. Select Save.
Private IP configuration	Select Dynamically allocate IP address.

Edit subnet network policy

To apply network security groups (NSGs) and user defined routes (UDRs) to a private endpoint, network policies must be enabled on the subnet. All private endpoints within the subnet are affected. [Learn more](#)

Virtual network: 8912-vnet1 (CST8912demo)
 Subnet: Subnet-1-8912 (10.0.0.0/24)

Network policies setting for all private endpoints in this subnet: **Enabled**

This change will affect all private endpoints in this subnet.

☒ Network security groups
☒ Route tables

Existing private endpoints

Name	Private IP	Resource	Sub-resource
------	------------	----------	--------------

Save **Discard**

3. Select Next: DNS.

Create a private endpoint

✓ Basics ✓ Resource ✓ Virtual Network **4 DNS** 5 Tags 6 Review + create

Private DNS integration

To connect privately with your private endpoint, you need a DNS record. We recommend that you integrate your private endpoint with a private DNS zone. You can also utilize your own DNS servers or create DNS records using the host files on your virtual machines. [Learn more](#)

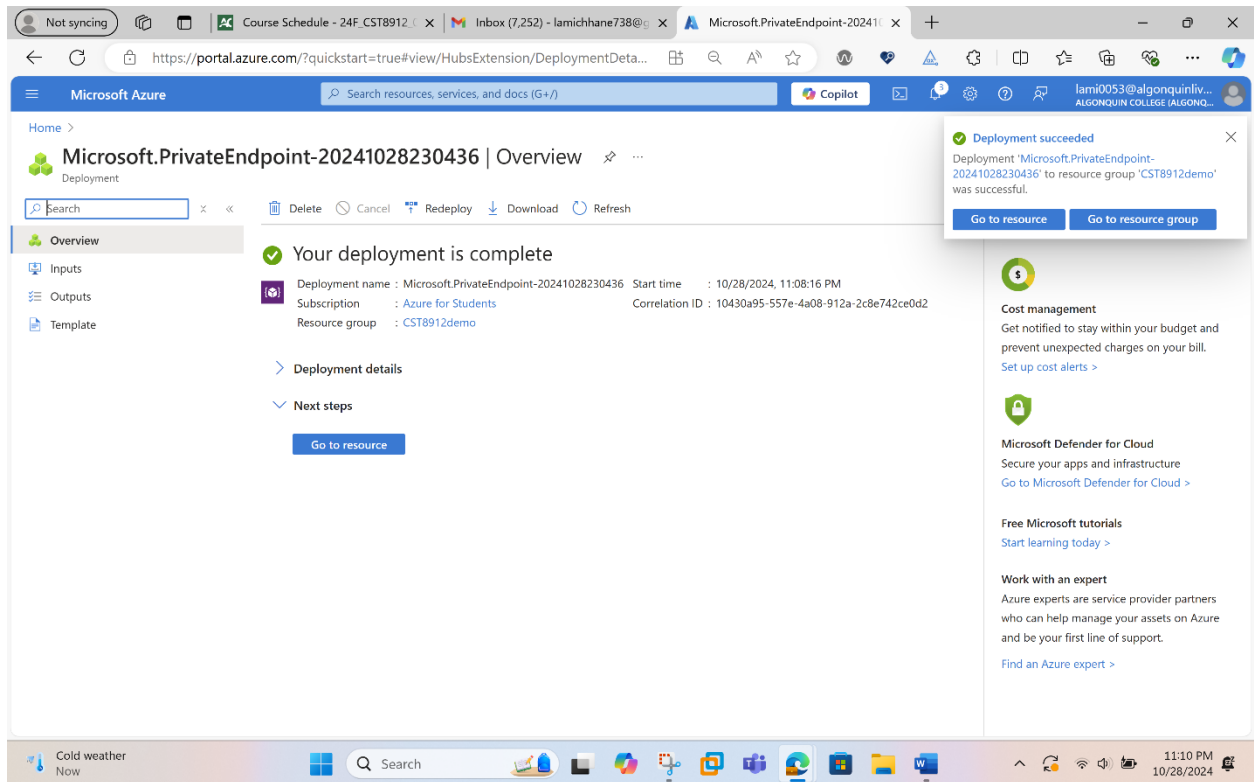
Integrate with private DNS zone: ☒ Yes ☐ No

Configuration name: privatelink-blob-core-win...
 Subscription: Azure for Students
 Resource group: CST8912demo
 Private DNS zone: (new) privatelink.blob.cor...

Next: Tags >

4.

5. Leave the defaults in DNS. Select Next: Tags, then Next: Review + create.
6. Select Create.

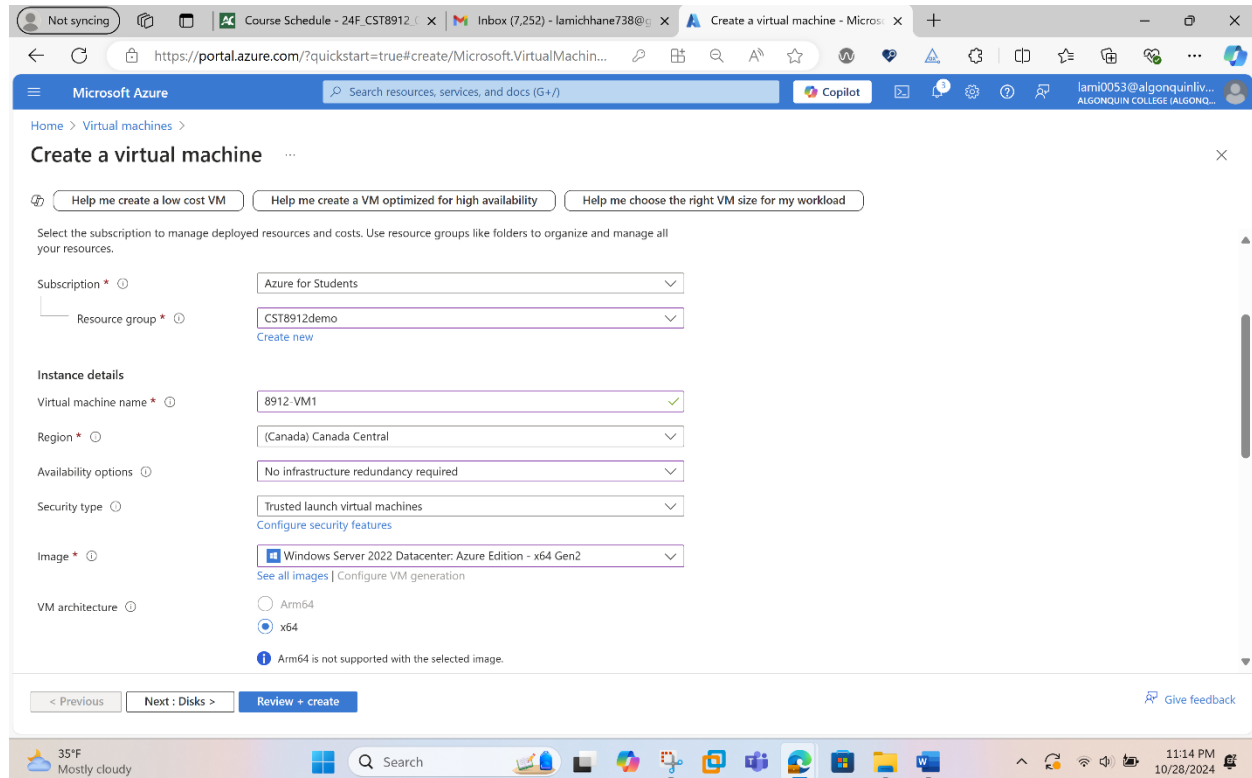


Step 5: Create test virtual machine

In the portal, search for and select Virtual machines. In Virtual machines, select + Create, then Azure virtual machine. On the Basics tab of Create a virtual machine, enter or select the following information:

Setting	Value
Subscription	Azure for students
Resource Group	CST8912demo
Virtual Machine name	8912-VM1
Region	Canada central
Availability options	No infrastructure redundancy required
Security type	Standard
Image	Select Windows Server 2022 Datacenter - x64 Gen2.

VM architecture	Leave the default of x64.
Size	Select size (B1)
Authentication	Choose username and password
Public inbound ports	Select None



Not syncing

Course Schedule - 24F_CST8912... | Inbox (7,252) - lamichhane738@... | Create a virtual machine - Micros...

https://portal.azure.com/?quickstart=true#create/Microsoft.VirtualMachin...

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

lami0053@algonquinliv... ALGONQUIN COLLEGE (ALGONQ...

Home > Virtual machines >

Create a virtual machine

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure for Students

Resource group * CST8912demo

Create new

Instance details

Virtual machine name * 8912-VM1

Region * (Canada) Canada Central

Availability options No infrastructure redundancy required

Security type Trusted launch virtual machines

Configure security features

Image * Windows Server 2022 Datacenter: Azure Edition - x64 Gen2

See all images | Configure VM generation

VM architecture

Arm64

x64

Arm64 is not supported with the selected image.

< Previous | Next : Disks > | Review + create

Give feedback

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Search

11:14 PM 10/28/2024

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Course Schedule - 24F_CST8912_C

Inbox (7,252) - lamichhane738@i

Create a virtual machine - Micros

https://portal.azure.com/?quickstart=true#create/Microsoft.VirtualMachin...

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

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Home > Virtual machines >

Create a virtual machine

Help me create a low cost VM Help me create a VM optimized for high availability Help me choose the right VM size for my workload

Size *

See all sizes

Enable Hibernation ☐

Hibernate is not supported by the size that you have selected. Choose a size that is compatible with Hibernation to enable this feature. [Learn more](#)

Administrator account

Username *

Password *

Confirm password *

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ☐ None ☒ Allow selected ports

Select inbound ports *

< Previous Next : Disks > Review + create

Give feedback

35°F Mostly cloudy

Search

11:15 PM 10/28/2024

Select the Networking tab at the top of the page. Enter the following information in the Networking tab:

Setting	Value
Virtual Network	Select 8912-vnet (CST8912demo).
Subnet	Select Subnet-1-8912.
Public IP	Select None
NIC network security group	Select Advanced
Configure network security group	Select Create new. Name it as 'nsg-1' for the name. Leave the rest at the defaults and select OK.

Not syncing | Course Schedule - 24F_CST8912... | Inbox (7,252) - lamichhane738@... | Create a virtual machine - Microsoft...

Microsoft Azure | Search resources, services, and docs (G+/I) | Copilot | lami0053@algonquiniv... ALGONQUIN COLLEGE (ALGONQ...

Home > Virtual machines >

Create a virtual machine

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Network interface
When creating a virtual machine, a network interface will be created for you.

Virtual network * [Create new](#)

Subnet * [Manage subnet configuration](#)

Public IP [Create new](#)

NIC network security group ☐ None ☐ Basic ☒ Advanced

Configure network security group * [Create new](#)

Delete NIC when VM is deleted ☐

Enable accelerated networking ☐ The selected VM size does not support accelerated networking.

Load balancers

< Previous | Next : Management > | Review + create

[Give feedback](#)

35°F Mostly cloudy | Search | 11:17 PM 10/28/2024

Now we will create a network security group.

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Microsoft Azure | Search resources, services, and docs (G+/I) | Copilot | lami0053@algonquiniv... ALGONQUIN COLLEGE (ALGONQ...

Home > Virtual machines >

Create network security group

Name * ✓

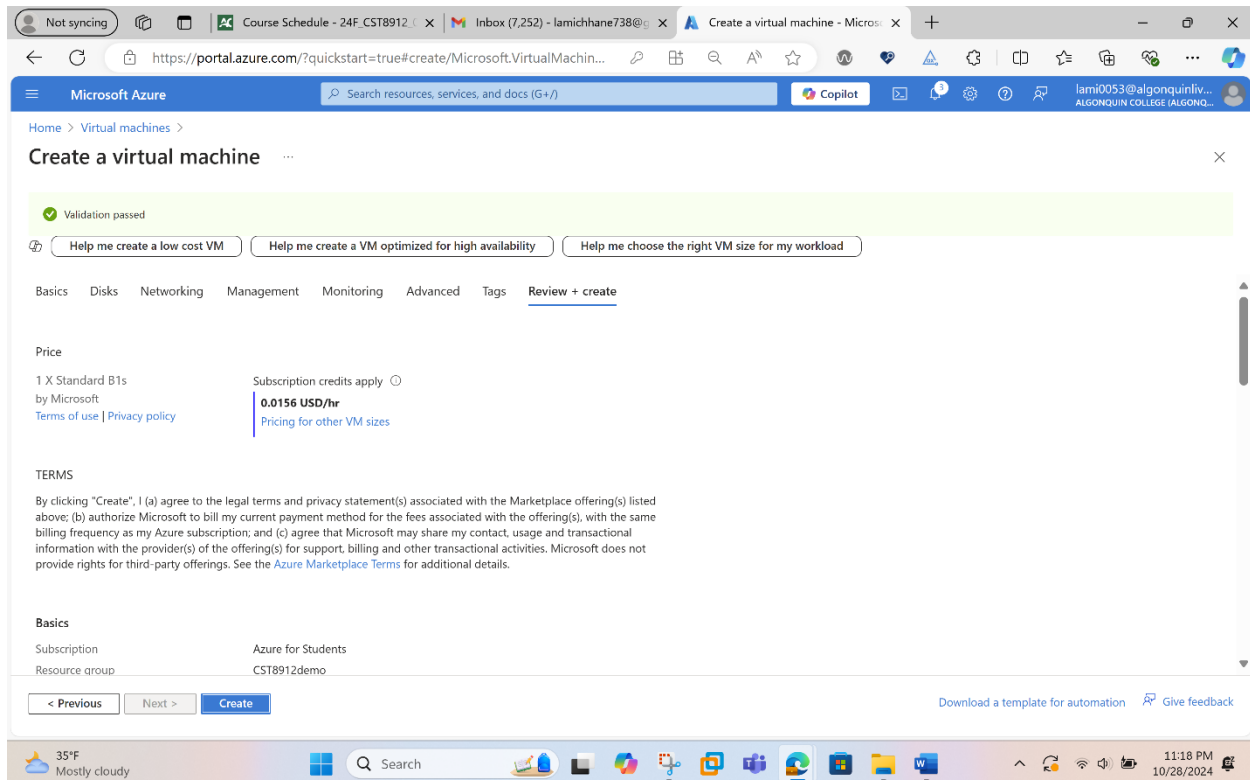
Inbound rules Any RDP (TCP/3389) ✓ [Add an inbound rule](#)

Outbound rules [Add an outbound rule](#)

OK

Upcoming Earnings | Search | 11:17 PM 10/28/2024

Leave the rest of the settings at the defaults and select Review + create. Review the settings and select Create.



Step 6: Storage access key

In the search box at the top of the portal, enter Storage account. Select Storage accounts in the search results. Then select the storage account we created in the previous steps or your existing storage account.

In the Security + networking section of the storage account, we select Access keys. And then we select Show, then select copy on the Connection string for key1.

Note: In this step copy the connection string, not the key.

The screenshot shows the Microsoft Azure portal interface. At the top, there's a navigation bar with the Microsoft Azure logo and a search bar. Below it, the breadcrumb trail indicates the path: Home > Storage accounts > storage48912. The main content area is titled 'storage48912 | Access keys'. It includes a search bar, a 'Set rotation reminder' button, a 'Refresh' button, and a 'Give feedback' button. The left sidebar shows a navigation menu with categories like 'Data storage', 'Security + networking', and 'Data management'. Under 'Data storage', 'Containers' is selected. The main content area displays two keys, key1 and key2, with their respective keys, connection strings, and rotation dates. The key1 key is highlighted, and its connection string is visible. The key2 key is also visible below it. The bottom of the screen shows a Windows taskbar with the search bar and various application icons.

Step 7 :Add a blob container

In the search box at the top of the portal, enter Storage account. Select Storage accounts in the search results. Select the storage account you created in the previous steps. In the Data storage section, select Containers. Select + Container to create a new container. Enter the container in Name and select Private (no anonymous access) under Public access level and finally select Create.

The screenshot displays the Microsoft Azure portal interface. The top navigation bar shows the user is logged in as 'lami0053@algonquinliv...'. The main content area is titled 'Storage accounts' and shows a list of storage accounts, with 'storage48912' selected. The 'Containers' section is expanded in the left sidebar. The 'New container' dialog is open on the right, showing the container name 'container' and the anonymous access level set to 'Private (no anonymous access)'. A message indicates that the access level is set to private because anonymous access is disabled on this storage account. The 'Create' button is visible at the bottom of the dialog.

The created container is shown below:

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and a user profile. The main content area is titled 'storage48912 | Containers'. On the left, a sidebar lists various services, with 'Containers' selected. The main panel displays a table of containers. A notification at the top right indicates that a storage container named 'container' has been successfully created.

Name	Last modified	Anonymous access le...	Lease state
☐ \$logs	10/28/2024, 11:02:52 ...	Private	Available
☐ container	10/28/2024, 11:22:08 ...	Private	Available

Step 8: Test connectivity to private endpoint

In the search box at the top of the portal, enter Virtual machine. Select Virtual machines in the search results.

- Select 8912-VM1.
- In Operations, select Bastion.

The screenshot shows the Microsoft Azure portal interface. At the top, there's a navigation bar with the Microsoft Azure logo and a search bar. Below it, the 'Virtual machines' section is visible, showing a list of virtual machines. The '8912-VM1' virtual machine is selected, and its details are displayed in the main pane. The 'Overview' tab is active, showing various properties of the virtual machine. The 'Connect' button is highlighted, and a dropdown menu is shown with options like 'Connect via Bastion'. The 'Properties' tab is also visible, showing details like 'Computer name', 'Operating system', 'Public IP address', and 'Networking'.

- Enter the username and password that we entered during the virtual machine creation.

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Not syncing | Course Schedule - 24F.CST8912.C | Inbox (7,252) - lamichhane738@ | 8912-VM1 - Microsoft Azure

https://portal.azure.com/?quickstart=true#@AlgonquinLivecom.onmicrosof...

Microsoft Azure | Search resources, services, and docs (G+/I) | Copilot | lami0053@algonquinliv... ALGONQUIN COLLEGE (ALGONQ...

Home > Virtual machines > 8912-VM1

Virtual machines

Algonquin College (AlgonquinLivecom.onmicrosof...

+ Create | Switch to classic | ...

Filter for any field...

Name ↑

8912-VM1

8912-VM1 | Bastion

Virtual machine

Search

Azure Bastion protects your virtual machines by secure and seamless RDP & SSH connectivity without the need to expose them through public IP addresses. [Learn more](#)

Using Bastion: **Bastion8912**

Provisioning State: **Succeeded**

Please enter username and password to your virtual machine to connect using Bastion.

Connection Settings

Keyboard Language ⓘ
English (US)

Authentication Type ⓘ
VM Password

Username ⓘ
keshav ✓

VM Password ⓘ
.....

Show

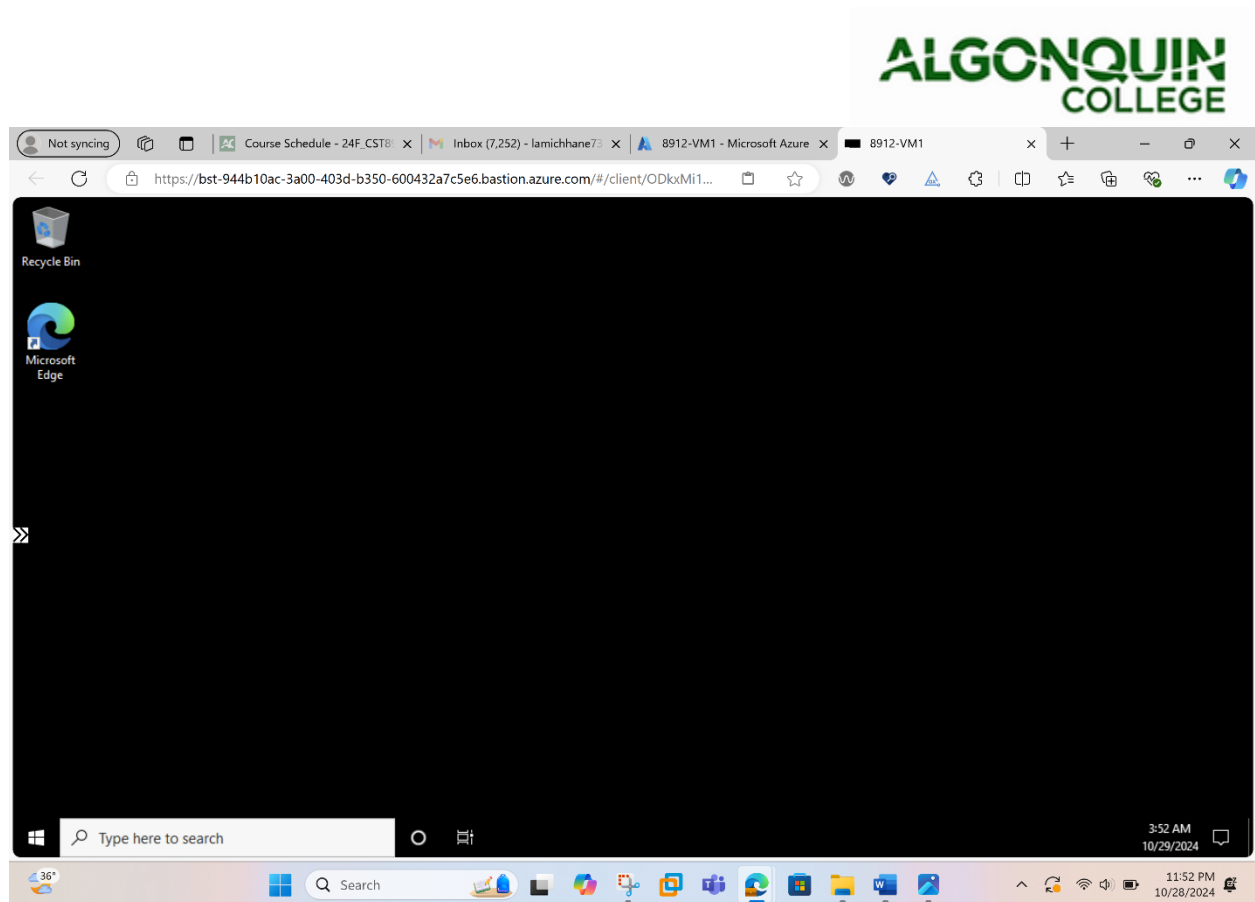
☒ Open in new browser tab

Connect

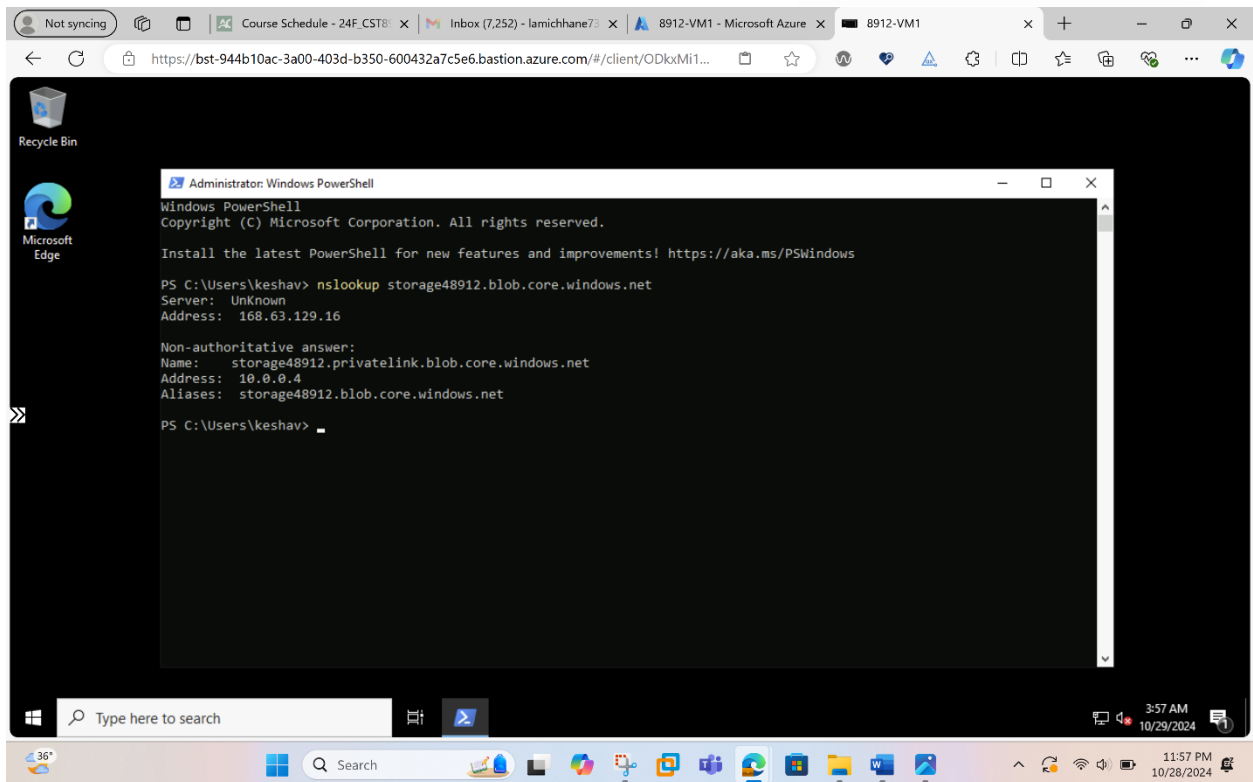
Tell us what you think of the Bastion experience

35°F Mostly cloudy | Search | 11:27 PM 10/28/2024

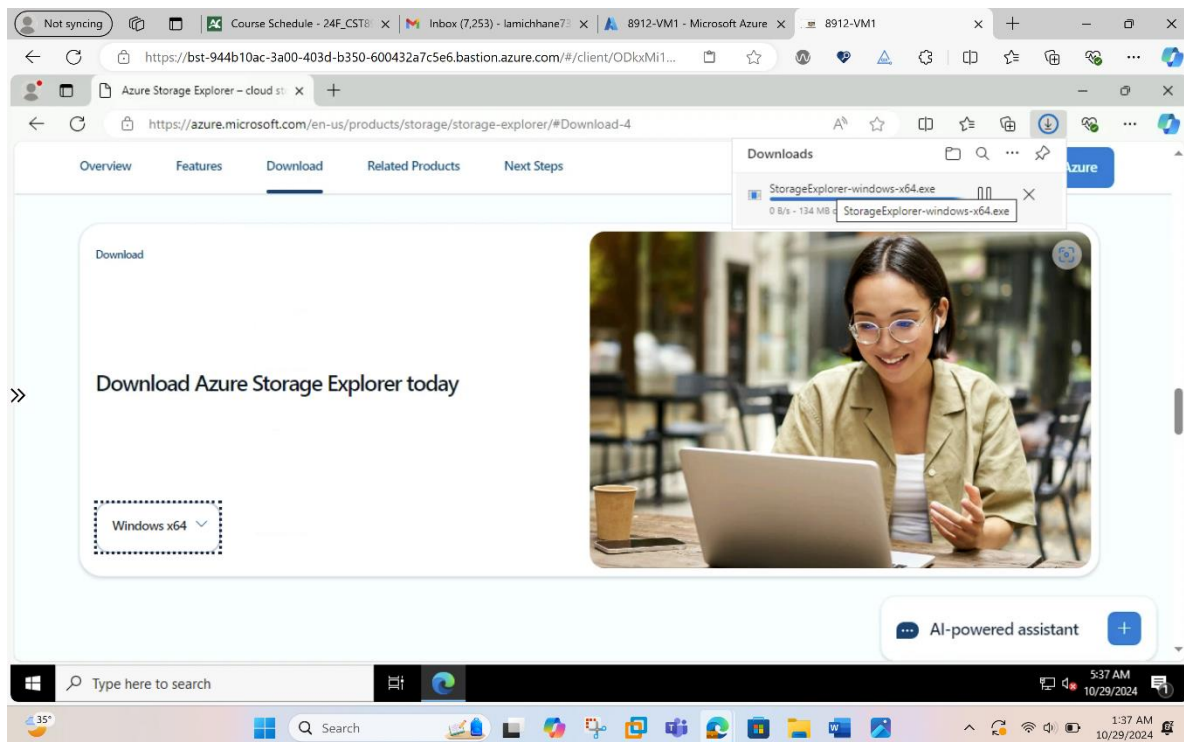
- Select Connect.



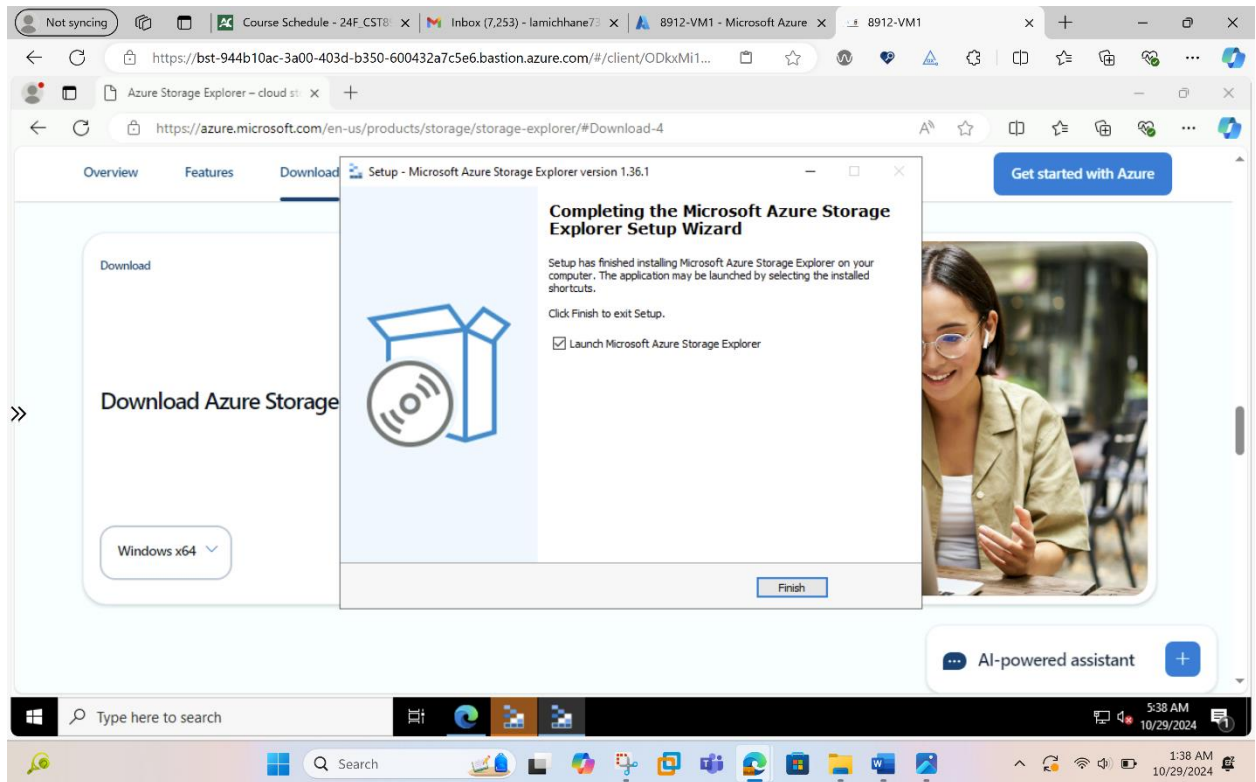
- Open Windows PowerShell on the server after you connect.
- Enter `nslookup <storage-account-name>.blob.core.windows.net`. Replace `<storage-account-name>` with the name of the storage account we created in the previous steps.



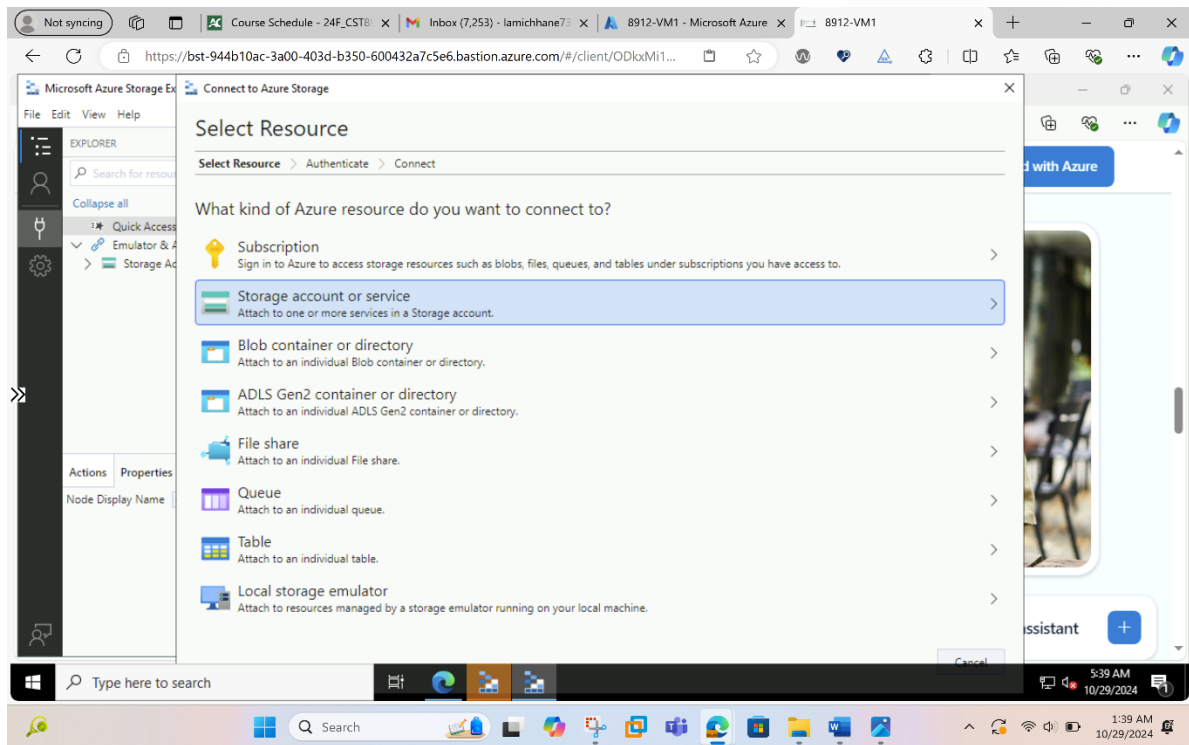
- Install Microsoft Azure Storage Explorer on the virtual machine.



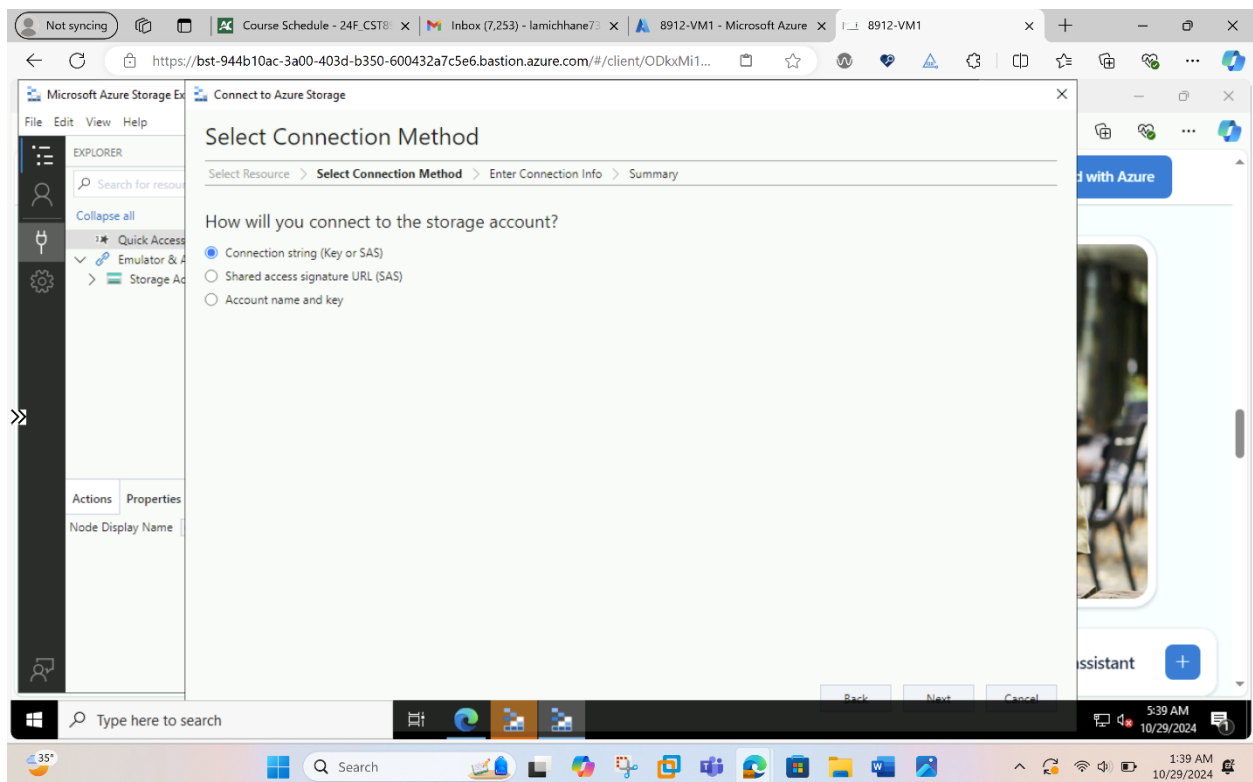
- Select Finish after the Microsoft Azure Storage Explorer is installed. Leave the box checked to open the application.



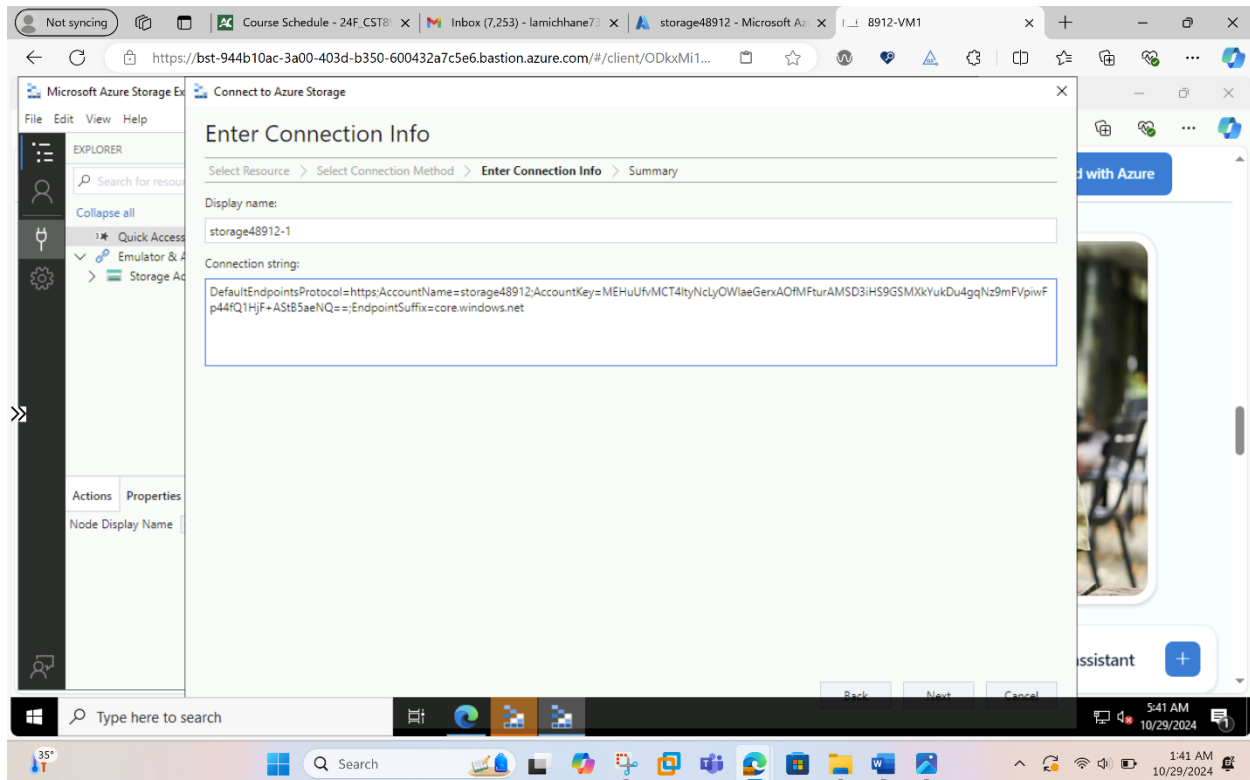
- Select the Power plug symbol to open the Select Resource dialog box in the left-hand toolbar.
- In Select Resource , select Storage account or service to add a connection in Microsoft Azure Storage Explorer to our storage account that we created in the previous steps.



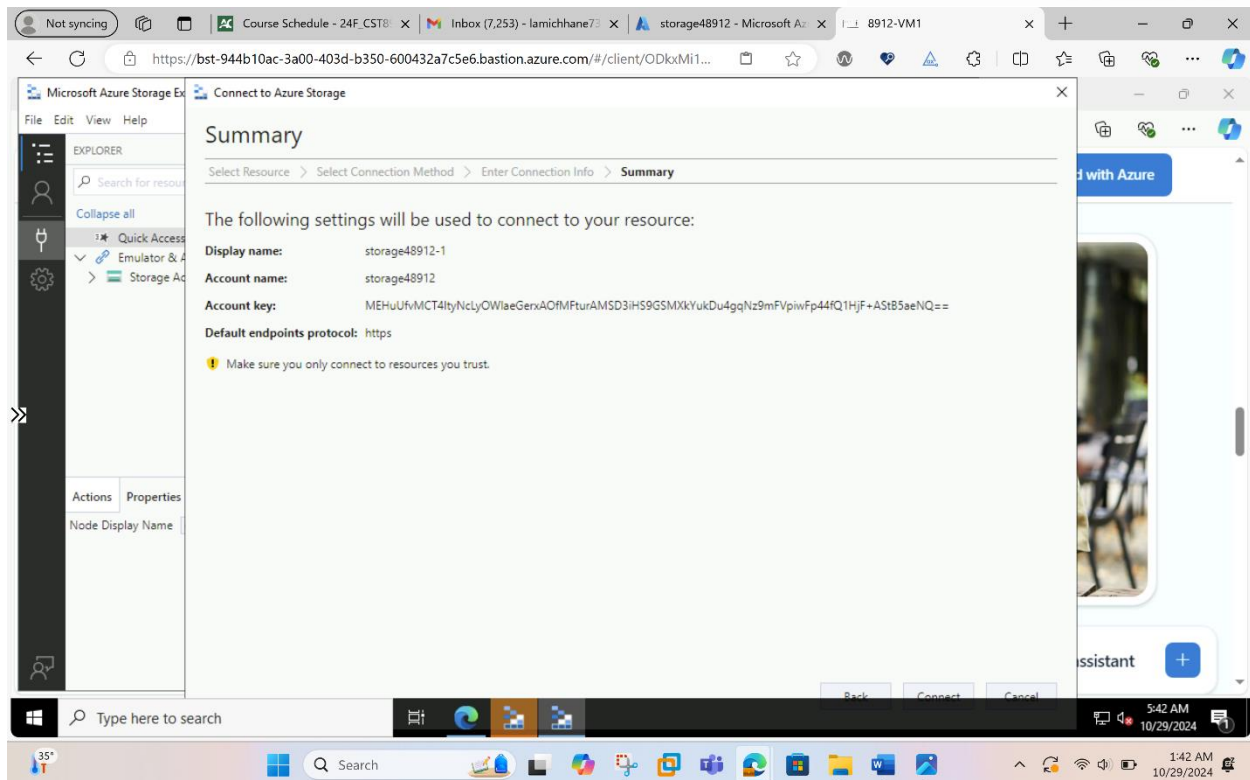
- In the Select Connection Method screen, select Connection string, and then Next.



- In the box under Connection String, paste the connection string from the storage account we copied in the previous steps. The storage account name automatically populates in the box under Display name.



- Select Next.
- Verify the settings are correct in Summary.



- Select Connect
 - ✓ Select the storage account from the Storage Accounts in the explorer menu.
 - ✓ Expand the storage account and then Blob Containers.
 - ✓ The container that we created previously is displayed.

The screenshot shows the Microsoft Azure Storage Explorer application. The left sidebar displays the Explorer view with the following structure:

- Quick Access
- Emulator & Attached
- Storage Accounts
 - (Attached Containers)
 - (Emulator - Default Ports)
 - storage48912-1 (Key)

The main pane shows the 'Get Started' tab for the 'storage48912-1 (Key)' account. It includes a 'Sign in with Azure' button and an 'Attach to a resource' button. The 'Resources' section on the right lists 'Documentation' and 'Troubleshooting Guide'. The 'Activities' pane at the bottom shows a successful connection message: 'Successfully added new connection.'

The Windows taskbar at the bottom shows the system clock as 5:42 AM on 10/29/2024.

The screenshot shows the Microsoft Azure Storage Explorer application with the 'storage48912' container selected. The left sidebar shows the Explorer view with the following structure:

- Quick Access
- Emulator & Attached
- Storage Accounts
 - (Attached Containers)
 - (Emulator - Default Ports) (Key)
 - storage48912-1 (Key)
 - Blob Containers
 - \$logs
 - container
 - File Shares
 - Queues
 - Tahlar

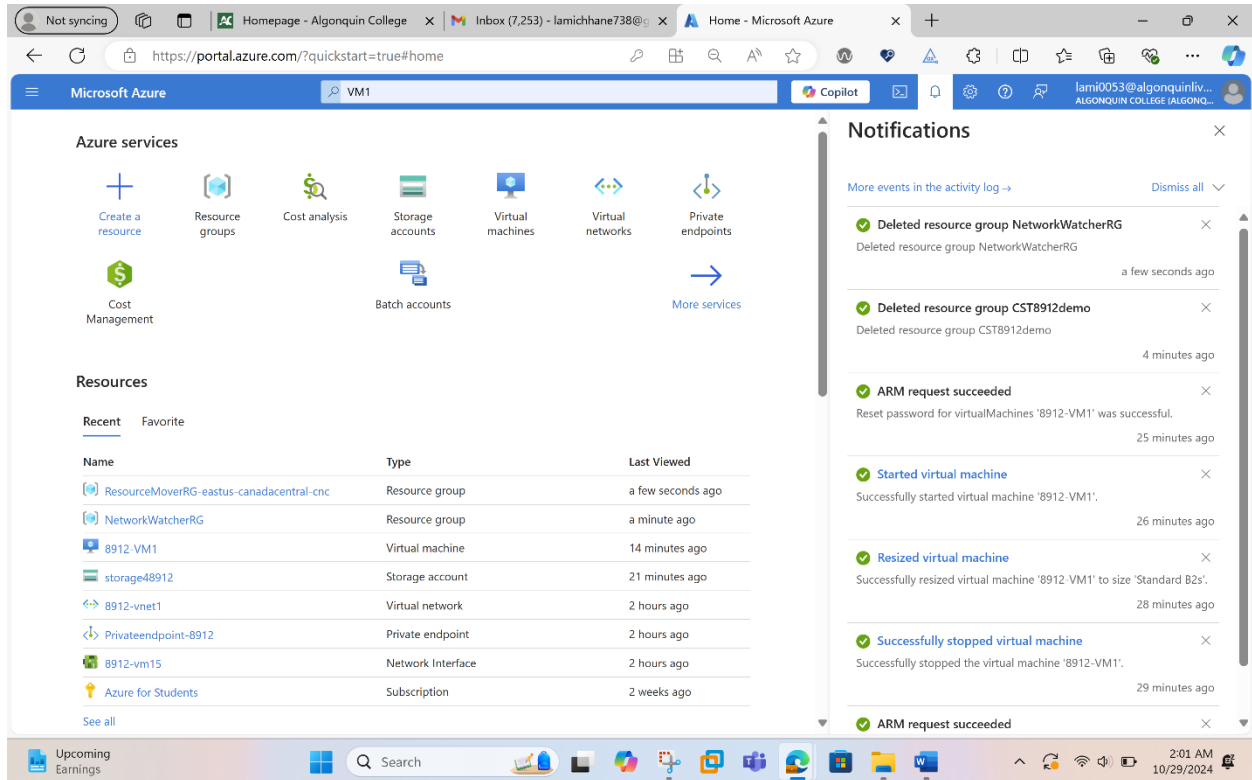
The main pane shows the 'storage48912' container view. It includes a 'Filter by prefix (case-sensitive)' search bar and a table of items:

Name	Last Modified	Public Access Level
\$logs	10/29/2024 3:02 AM	off
container	10/29/2024 3:22 AM	off

The 'Activities' pane at the bottom shows a successful connection message: 'Successfully added new connection.'

The Windows taskbar at the bottom shows the system clock as 5:42 AM on 10/29/2024.

- Close the connection to 8912-VM1.
- Clean all the resources created during the lab .



The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the 'Microsoft Azure' logo, a search bar with 'VM1' entered, and a user profile for 'lami0053@algonquinliv...'. The main content area is divided into two sections: 'Azure services' and 'Resources'.

Azure services section includes icons for 'Create a resource', 'Resource groups', 'Cost analysis', 'Storage accounts', 'Virtual machines', 'Virtual networks', 'Private endpoints', 'Cost Management', and 'Batch accounts'.

Resources section shows a table of recent resources:

Name	Type	Last Viewed
ResourceMoverRG-eastus-canadacentral-cnc	Resource group	a few seconds ago
NetworkWatcherRG	Resource group	a minute ago
8912-VM1	Virtual machine	14 minutes ago
storage48912	Storage account	21 minutes ago
8912-vnet1	Virtual network	2 hours ago
Privateendpoint-8912	Private endpoint	2 hours ago
8912-vm15	Network Interface	2 hours ago
Azure for Students	Subscription	2 weeks ago

On the right side, a 'Notifications' panel displays a list of events:

- Deleted resource group NetworkWatcherRG (a few seconds ago)
- Deleted resource group CST8912demo (4 minutes ago)
- ARM request succeeded: Reset password for virtualMachines '8912-VM1' was successful. (25 minutes ago)
- Started virtual machine: Successfully started virtual machine '8912-VM1'. (26 minutes ago)
- Resized virtual machine: Successfully resized virtual machine '8912-VM1' to size 'Standard B2s'. (28 minutes ago)
- Successfully stopped virtual machine: Successfully stopped the virtual machine '8912-VM1'. (29 minutes ago)
- ARM request succeeded

This final screenshot ensures that all the resources created for this lab were deleted successfully as per the requirement of lab instruction.

Conclusion:

In this lab, we successfully set up Azure Private Link and Private Endpoint, guaranteeing private and secure access to Azure services over a virtual network. This exercise taught us how to join Azure resources securely without exposing data to the public internet by configuring a private endpoint, turning off public access, and testing connectivity. This configuration provides a strong basis for a secure network design in Azure by improving data privacy, satisfying compliance requirements, and lowering the risks related to internet exposure.